CN_Module5_MidpointReport_Group07

Team Name/Members:

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Project Title: Analyzing the network latency of High Frequency Trading.

1. Motivation
   - **Industry Trend:** Modern financial markets have shifted toward algorithmic execution, where **High-Frequency Trading (HFT)** represents the cutting edge of network performance requirements.
   - **Real-World Relevance:** In HFT environments, a latency difference of a single millisecond (ms) or microsecond () directly translates into millions of dollars in potential profit or loss. Therefore, optimizing the physical distance and protocol efficiency of the network is a matter of survival for financial institutions.
   - **Job-Ready Value:** Beyond basic coding, the ability to **diagnose protocol-level bottlenecks** and explain the "Why & How" of network delays is a highly sought-after skill in the fintech and infrastructure sectors.
2. Problem Definition
   - **The Core Challenge:** Traditional network architectures suffer from **propagation delay ()** and **queuing delay ()**, which act as "fatal weaknesses" in HFT. Specifically, as traffic intensity () approaches 1 during high market volatility, queuing delays increase exponentially, leading to catastrophic execution lag or packet loss.
   - **Scope of Exploration:**
     - Analyzing the trade-offs between **TCP (reliability)** and **UDP (speed)** in HFT socket communication, focusing on the overhead of TCP's 3-way handshake and congestion control.
     - Investigating strategies to minimize physical distance through **Co-location** and the use of high-speed transmission media such as **Microwave networks**.
   - **Target Outcome:** A professional **Tech Blog (GitHub README)** that provides a mathematical breakdown of the four sources of delay () and evaluates the effectiveness of current industry solutions.
3. Plan and Timeline
   - **Methodology:** Literature review of **IEEE papers** and **IETF RFCs**, combined with case study analysis of leading financial technology providers (e.g., Cisco, AWS).
   - **Milestones:**
     - **Week 11:** Initial research and gathering of credible references (IEEE papers, industry whitepapers).
     - **Week 12:** Deep-dive analysis of transport layer protocols (TCP/UDP/QUIC) and mathematical modeling of delay factors.
     - **Week 13:** Comparative analysis of hardware-level optimizations (FPGA, Microwave links) and

data synthesis.

- **Week 14:** Finalizing the **GitHub Tech Blog** documentation and recording the English presentation video.

4. Individual Roles

- 손한주 Son Hanju : Simulation lead(Test environment setup Benchmark code skeleton)
- 이준서 Lee Junseo: Industry lead(Company case studies Trade-off comparison table)
- 김보석 Kim Boseok: Problem lead(HTF domai, fatal weakness, Midpoint report draft)
- 김정하 Kim Jungha: Protocal lead(DPDK, OpenOnlad, RDMA Protocl-level analysis)
- (Ensure responsibilities are evenly distributed and clearly defined).
- Who will be responsible for record?
  : we will record with together in turns.

5. References (if you find already)

[1] J. Lockwood et al., "A Low-Latency Library in FPGA Hardware for High-Frequency Trading (HFT)," *IEEE Symposium on High-Performance Interconnects*, 2012,
https://ieeexplore.ieee.org/document/6299067
[2] J. Postel, "Transmission Control Protocol," IETF RFC 793, September 1981.
https://datatracker.ietf.org/doc/html/rfc793
[3] G. Laughlin, A. Aguirre, J. Grundfest, "Information Transmission Between Financial Markets in Chicago and New York," *Financial Review*, 2014.
https://arxiv.org/abs/1302.5966
[4] M. Aquilina, E. Budish, P. O'Neill, "Quantifying the High-Frequency Trading 'Arms Race'," *The Quarterly Journal of Economics*, Vol. 137, No. 1, 2022.
https://academic.oup.com/qje/article/137/1/493/6368348
[5] D. Bhattacherjee et al., "A Bird's Eye View of the World's Fastest Networks," *ACM Internet Measurement Conference (IMC)*, 2020.
https://bdebopam.github.io/papers/imc2020-hft.pdf